Course: Operating Systems Architecture
Unit 1: Process Management & CPU Scheduling
An operating system manages concurrent Process Scheduling using priority queues. Context switching saves the execution state of a thread to resume it later.

Unit 2: Concurrency & Synchronization

Thread Synchronization prevents race conditions when accessing shared memory regio

Mutex Locks ensure mutual exclusion within the critical section of concurrent code.

Unit 3: Deadlock Detection & Prevention

Deadlock occurs when processes are blocked waiting for circular resource allocation.

The Banker’s Algorithm ensures safe system states by evaluating maximum resource d